MIL-T-13333B(MU)
28 May 1965
SUPERSEDING
MIL-T-13333A(Ord)
1 September 1959

MILITARY SPECIFICATION

TELESCOPE, M49

1. SCOPE

1.1 This specification covers one type of 20-power telescope for observing effectiveness of artillery or small arms fire.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

U.S. Army Munitions Command

F7578445	Telescope, M49
IEL 7578445	List of Inspection Documents for Telescope M49

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with Drawing F7578445 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental service conditions.

3.3.1 Storage temperatures.- The telescope shall show no evidence of physical failure such as glass breakage, bonding failure, or casting failure due to storage at temperatures from -80° to +160° Fahrenheit (F).

3.3.2 Standard ambient temperature.- The telescope shall meet the requirements of 3.4 through 3.5.3 at a standard ambient temperature of 75° ± 15°F.

3.3.3 Vibration.- The telescope shall be vibrated at a constant frequency of 30 ± 5 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion). Subsequent to vibration there shall be no evidence of damage or physical failure and the telescope shall meet the requirements of 3.4 to 3.5.3 inclusive.

3.4 Cleanliness.- Subsequent to vibration the telescope shall meet the cleanliness requirements of MIL-C-13830. There shall be no more than 2 particles of dirt visible when viewed through the eyepiece. The area of each particle shall not be in excess of the area equivalent to a size 50 dig.

3.5 Accuracy.

3.5.1 Resolution.- The resolving power on the geometric axis shall be 3½ seconds of arc or less.

3.5.2 Image focus.- Rotation of the focus adjustment shall bring a target into sharp focus at any distance between 50 feet and infinity.

3.5.3 Image tilt.- The image of a plumb line shall be vertical within 45 minutes of arc.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of Product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 7578445. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with the "Classification of Defects" specified in Table I below and the inspection level and sampling plans specified in MIL-STD-105. Examinations and tests for Packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

CRITICAL: NONE	REQUIREMENTS	TEST PROCEDURES
MAJOR: AQL 0.65 Percent Defective		
101. Vibration	3.3.3	4.5.2.1
102. Cleanliness	3.4	4.5.2.2
103. Resolution	3.5.1	4.5.2.4
104. Image focus	3.5.2	4.5.2.5
105. Image tilt	3.5.3	4.5.2.6
MINOR: None		

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4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Special sampling.

4.3.1 General.- One telescope shall be selected at random by a Government representative as a sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specification	3.2	MIL-F-13926 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as samples from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table I and shall then meet the requirement and test in Table III.

TABLE III

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
303. Storage temperatures	3.3.1	4.5.1.1

4.3.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents, IEL 7578445.

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4.4.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.3 Utilization of standard test equipment.- The test equipment used to perform the optical test shall consist of a standard shop type collimator projector having provisions for setting target distances between fifty feet and infinity. This equipment shall also provide a supporting medium to position the telescope under test and a dioptometer behind the eyepiece of the telescope under test. Provisions shall also be provided for the alignment of the collimator projector's optical axis and the optical axis of the dioptometer to the proposed optical axis of the telescope when positioned in the holding medium. The collimator projectors reticle shall consist of a horizontal and vertical reticle line. The intersecting point of these lines shall fall on the optical axis of the collimator. Resolution test chart representing the resolving power of the telescope under test (see 3.5.1) shall be constructed about this intersecting point as shown in Figure 1 of MIL-O-13830. Angular tolerance lines shall be located each side of the vertical line above and below the horizontal reticle line to represent the image tilt limit (see 3.5.3). The location and spacing of all reticle lines, test bars, target distances, shall be positioned within the tolerance specified in 4.4.2.2.

4.4.3.1 Control of environmental test equipment.- In conjunction with 4.4.2.2 the following test equipment shall be utilized in the performance of the applicable tests specified in 4.5:

Item Number	Equipment	Test Requirements
1	Heat Chamber	Test facilities of MIL-F-13926
2	Cold Chamber	Test facilities of MIL-F-13926
3	Vibration fixture	Test facilities of MIL-F-13926

4.5 Test methods and procedures.

4.5.1 Environmental

4.5.1.1 Storage temperature.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The telescope shall be exposed

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to each temperature extreme until thermal stabilization is reached. After each temperature exposure, the telescope shall be stabilized in an ambient temperature between +60° and 90°F and subjected to an audible, tactile and visual examination. The telescope shall conform to the requirements of 3.3.1. Caution should be exercised during this test to avoid subjecting the telescope to thermal shock.

4.5.2 Performance.

4.5.2.1 Vibration.- The telescope shall be mounted in its normal operating position to the test equipment by means of an adapter so that the entire unit is immobilized. The telescope shall be vibrated in a vertical plane with the specified motion and frequency applied for 5 minutes ± 15 seconds. Subsequent to vibration the telescope shall be visually examined for missing or damaged parts. A tactile examination should also be conducted to determine looseness or improper function. An audible examination shall be made by positioning the telescope close to the ear and vigorously shaking it with both hands while listening for loose internal optical and mechanical parts. Any one of the possible defects defined above shall be cause to consider a telescope defective. Telescopes which have passed the above examinations shall be subjected to the tests as outlined below and shall conform to the requirements specified therein.

4.5.2.2 Cleanliness.- The optical surfaces shall be inspected for cleanliness in accordance with MIL-O-13830. The telescope shall conform to the cleanliness requirements of 3.4.

4.5.2.3 Test equipment set-up.- The optical axis of the collimator projector and of the dioptometer shall be aligned to the holding medium used to support the telescope under test, by means of auto-collimation. The vertical reticle lines of the projector collimator and of the dioptometer shall be positioned parallel to a plumb line. The dioptometer shall be set to proper focus by directing the line of sight toward a white background and adjusting the eye assembly to produce a sharp image of the reticle. Parallax shall be removed by setting the objective scale to zero. Parallax is defined as movement between the selected target point and the intersecting point of the horizontal and vertical reticle line when the observers head is moved side to side or up and down.

4.5.2.4 Resolution.- This test is performed with the test equipment aligned as outlined in 4.5.2.3. Set the collimator projector to minimum target distance and sight through the telescope to obtain a sharp image of the vertical and horizontal reticle lines of the collimator. This image is obtained by adjusting the telescopes eyepiece. During this adjustment the eyepiece assembly shall not contact the stop as specified in 3.5.2. Position the dioptometer behind the telescopes eyepiece and observe the test bar pattern (see 4.4.3). They shall be resolved in the four quadrants within the limits outlined in 3.5.1 as measured by means of

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the diopter scale of the dioptometer. Resolution is assured when the observer can determine the direction in which the lines lay. Repeat this procedure with the collimator projector and telescope set to infinity.

4.5.2.5 Image focus.- This test is performed in conjunction with the resolution test as outlined above. When proper image focus has been obtained as specified in 3.5.2 the eyepiece assembly shall not contact the stop when using minimum or maximum target distances.

4.5.2.6 Image tilt.- Using the set-up outlined in 4.5.2.3 the test for image tilt is performed by sighting through the dioptometer and observing parallelism between the vertical reticle line of the projector collimator and the dioptometer. Both lines must be parallel within the limits of the tolerance lines of the collimator projector reticle to conform with the requirement specified in 3.5.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Unless otherwise specified packaging and packing shall be in accordance with Level C of MIL-P-14232. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The Telescope M49 is primarily intended for observation of artillery or small arm fire. This telescope is a 20 power scope without a reticle. The telescope is supported on Tripod: M15 and is carried in Case Carrying: M164. The tripod: M15 is carried in Case Carrying: M42.

6.2 Ordering data.- Procurement should specify the following:

(a) Title, number, and date of this specification
(b) Applicable packaging data (see 5.1)

Custodian:

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Preparing activity:

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